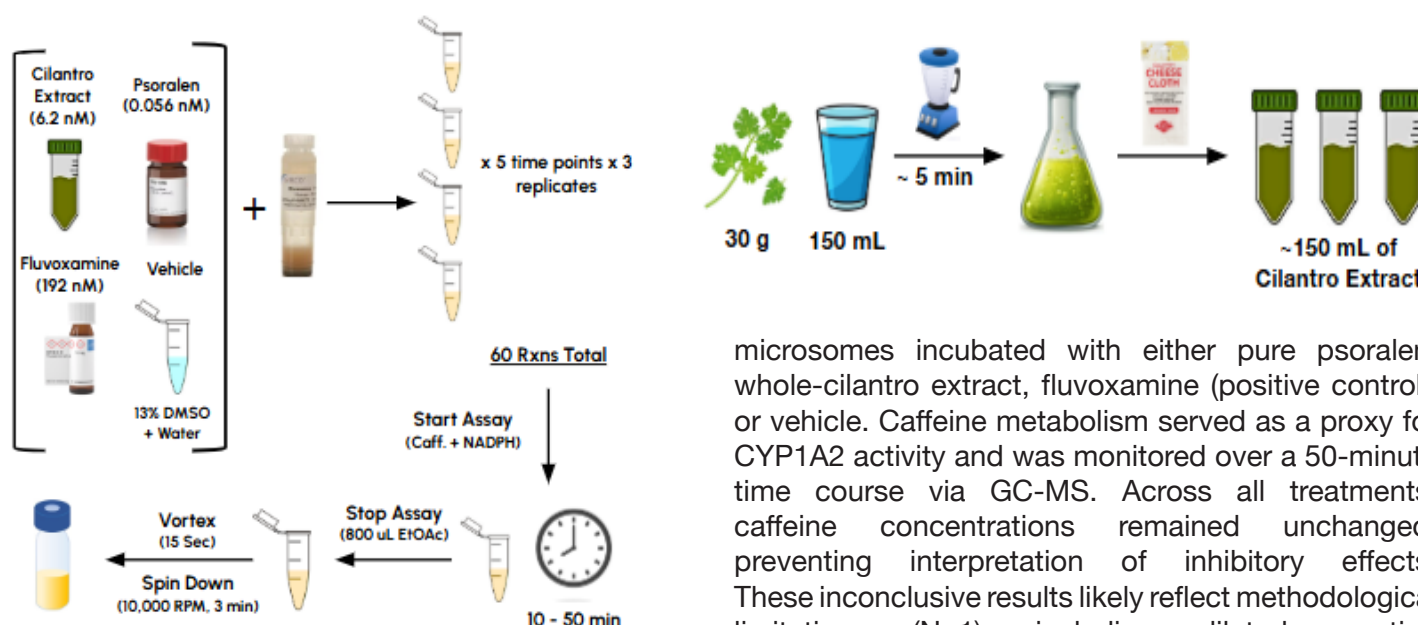


Evaluating Natural CYP1A2 Inhibitors: Psoralen Quantification and Comparative Inhibition by Pure Compound and Cilantro Extract in Human Liver Microsomes

Chris Flores



Abstract:

Cytochrome P450 1A2 (CYP1A2) plays a key role in xenobiotic metabolism and is responsible for approximately 95% of caffeine clearance in humans. Naturally occurring inhibitors of CYP1A2, including furanocoumarins found in Apiaceae vegetables, alter drug metabolism and contribute to clinically relevant food-drug interactions. To investigate whether whole-plant extracts exhibit greater inhibitory activity than isolated compounds, we compared CYP1A2 inhibition by pure psoralen with that of a cilantro extract containing multiple naturally occurring phytochemicals. Additionally, we quantified psoralen abundance in cilantro using a newly developed GC-MS method. Psoralen was successfully detected using selected ion monitoring (m/z 187), and a standard curve confirmed linear quantification. GC-MS analysis revealed 716.7 ng/g psoralen in fresh cilantro, a concentration higher than previously reported values. CYP1A2 inhibition was assessed using human liver

microsomes incubated with either pure psoralen, whole-cilantro extract, fluvoxamine (positive control), or vehicle. Caffeine metabolism served as a proxy for CYP1A2 activity and was monitored over a 50-minute time course via GC-MS. Across all treatments, caffeine concentrations remained unchanged, preventing interpretation of inhibitory effects. These inconclusive results likely reflect methodological limitations ($N=1$), including diluted reaction components, suboptimal assay temperature, and insufficient turnover time. Despite this, the study established a functional psoralen quantification workflow and provides a framework for refinement of microsomal assays. Future studies should optimize reaction conditions, test additional furanocoumarins, and validate CYP1A2 inhibition at shorter, single time points before full time-course analysis.

Keywords: CYP1A2 Inhibition • Furanocoumarins • Human Liver Microsomes • Psoralen Quantification • Food-drug Interactions

Introduction

Cytochrome P450s are a large class of heme-containing enzymes responsible for the metabolism of drugs and other xenobiotics (Basic Review of the Cytochrome P450 System, 2013). Among them, CYP1A2, a member of the CYP1A subfamily, plays a major role in drug metabolism (Thorn et al., 2012).

CYP1A2 is estimated to account for approximately 95% of caffeine metabolism in humans (Cornelis et al., 2006) and is known to metabolize over one hundred additional substrates (Thorn et al., 2012), a consequence of its relatively promiscuous and flexible binding pocket. In addition to xenobiotics, CYP1A2 metabolizes many endogenous compounds such as steroids and arachidonic acid (Zhou et al., 2009). CYP1A2 is expressed primarily in liver tissue (Hwang et al., 2021; Thorn et al., 2012), where it functions by coupling the oxidation of NADPH, via cytochrome P450 reductase, with the oxidation of a substrate. This chemical change makes the substrate, often a drug, more polar, thus increasing clearance.

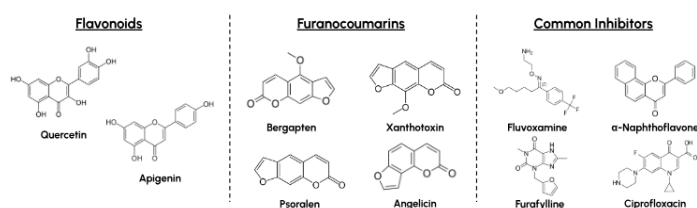


Figure 1. Inhibitors of CYP1A2. Family groups of natural inhibitors found in Apiaceous vegetables, such as flavonoids and furanocoumarins, and common synthetic inhibitors.

The enzymatic activity of CYP1A2 can be inhibited by many different compounds, many of which compete with substrates for access to the active site or bind allosterically to alter enzyme conformation, therefore altering substrate affinity. Several inhibitors have been identified and well-characterized for CYP1A2. Synthetic inhibitors such as fluvoxamine, α -naphthoflavone, furaflavone, and ciprofloxacin are frequently used in pharmacological studies to characterize enzyme kinetics and confirm isoform specificity (Bourri  et al., 1996; Liu et al., 2013) (Figure 1). In addition to synthetic xenobiotics, naturally occurring inhibitors of CYP1A2 include plant-derived furanocoumarins (e.g., psoralen, bergapten, xanthotoxin) and flavonoids (e.g., quercetin, apigenin), which are commonly found in Apiaceae vegetables (Alehaideb et al., 2021; Berteina-Raboin, 2025; Kang et al., 2011). Despite their known inhibitory activity, literature on furanocoumarins remains limited, which poses a significant health concern as CYP1A2 metabolizes many therapeutics, including caffeine, clozapine, and other aromatic drugs. Thus, CYP1A2 inhibition by dietary furanocoumarins from Apiaceae vegetables, such as celery, parsley, parsnip, and cilantro, may alter drug clearance and plasma concentrations, emphasizing the clinical importance of understanding these food-drug interactions (Alehaideb, 2023; Kang et al., 2011).

Intrigued by the range of natural inhibitors, we sought to investigate whether psoralen, the parent molecule of many furanocoumarins, and other natural CYP1A2 inhibitors exhibit cooperative effects when consumed together within an Apiaceae vegetable. To assess this potential cooperativity, we compared CYP1A2 inhibition by a whole-plant cilantro extract with that of pure psoralen. We hypothesized that treatment with the cilantro extract would produce stronger inhibition than the pure psoralen compound, as the whole-plant extract likely contains multiple inhibitory compounds that function as a naturally occurring inhibitory cocktail. To evaluate potential synergistic interactions, we compared the magnitude of CYP1A2 inhibition between the cilantro extract and commercial psoralen. To compare CYP1A2 inhibition, human liver microsomes were selected as the *in vitro* system for CYP1A2 activity assays. Microsomes are small vesicles (100-200 nm) that contain concentrated amounts of cytochrome P450 enzymes (Alberts et al., 2002). Although microsomes are not naturally occurring cellular structures, they form through the fragmentation and resealing of the endoplasmic reticulum (ER) after disruption by sonication and centrifugation (Appendix A). Due to their enrichment in ER-derived P450s, microsomes are commonly used as model systems to characterize the metabolic activity of cytochrome P450 enzyme variants (Alehaideb et al., 2021). Given the limited literature on psoralen abundance in cilantro, we also quantified psoralen levels by GC-MS and compared our measured concentration (ng/g fresh weight) with current literature values. These findings may guide future bioprospecting efforts for naturally occurring multi-compound inhibitors and additional furanocoumarin derivatives. Furthermore, elucidating the active inhibitory components of cilantro will also contribute to a deeper understanding of diet-mediated food-drug interactions involving CYP1A2.

Methods

Experiments were carried out following the "Independent Project Week 1 - Psoralen Standard Curve and Cilantro Extraction" and "Independent Project Week 2 - Microsomal Assays, Caffeine Extractions, and Second Cilantro Extraction" protocols from the BIOL 303 Pharmacology lab component, kindly provided by Cynthia Holland, Ph.D. (Department of Biology, Williams College). Final concentrations and volumes are listed in greater detail below. Preparation of the psoralen standard curve. A standard curve relating psoralen concentration ($\mu\text{g/mL}$) to GC-MS peak area was generated by preparing

a series of psoralen standards (0.01, 0.1, 1.0, 10, and 100 µg/mL in EtOAc; Cayman Chemical). Each standard was analyzed by GC-MS using a newly developed SIM method (2025_HollandLab_Psoralen_Splitless_SIM_8min.M) adapted from a previous caffeine protocol (2024_HollandLab_Caffeine_Splitless_SIM_8min.M). Peak areas were recorded and plotted against psoralen concentration, and a linear regression was performed on GraphPad Prism (version 10.4.1) to construct the standard curve.

Preparation of stock cilantro extract. A cilantro extract was prepared by blending raw cilantro (30g; Stop&Shop, Williamstown, MA) with dH₂O (150 mL). The aqueous extract was then decanted and filtered to remove particulates. A fraction (45 mL) of the cilantro extract was reserved for GC-MS analysis, and the remaining extract was wrapped in aluminum foil and stored at -20 °C until it was thawed for use in microsomal assays.

Psoralen extraction from cilantro extract. Psoralen was extracted by combining 5 mL of EtOAc with 45 mL of cilantro extract. The mixture was vortexed for 30 seconds at max speed and spun down at 10,000 RPM for 3 minutes. Following phase separation, 600 µL of the top EtOAc layer was transferred to a labeled GC-MS vial for analysis.

Psoralen GC-MS method. Psoralen extracted from the cilantro extract was analyzed by GC-MS using the method 2025_HollandLab_Psoralen_Splitless_SIM_8min. Samples were run in selected ion monitoring (SIM) mode at m/z 187, corresponding to the protonated [M+1] molecular ion of psoralen. Helium served as the carrier gas at a constant flow rate of 1.9 mL/min. Injections were made in split mode (10:1). The oven was programmed to initially hold for 2 min at 60 °C, followed by a ramp to 300 °C at 60 °C/min, and finally hold at 300 °C for 2 min. The transfer line temperature was maintained at 280 °C. The mass spectrometer was operated with an electron ionization (EI) source at 70 eV with a source temperature of 180 °C.

Timed microsomal assays. Human liver microsomes (50 donors; ThermoFisher Scientific) expressing CYP1A2 were incubated with their respective inhibitor treatments for 15 minutes prior to initiating the assays. Each reaction consisted of 5 µL microsomes (20 mg/mL), 183 µL of 10 mM PBS (pH 7.4; ThermoFisher Scientific), and 2 µL of either of the following: pure psoralen (0.056 nM; 0.14% DMSO in dH₂O), cilantro extract (psoralen concentration: 6.2 nM), fluvoxamine (192 nM in 10 mM PBS, pH 7.4), or vehicle (0.14% DMSO in dH₂O). Reaction mixtures were incubated at room temperature. Treatment with the vehicle served as the negative control, to simulate

no CYP1A2 inhibition, whereas the fluvoxamine served as a positive control, maximizing inhibition. To measure CYP1A2 activity, caffeine was added to each reaction (final assay concentration: 0.05 mM). The metabolism of caffeine during the assay was measured to assess CYP1A2 activity in the presence of different inhibitors. Timed assays (10, 20, 30, 40, and 50 minutes) were conducted in triplicate. Staggering the start of each reaction by one minute, reactions were initiated with 10 µL of NADPH (9.23 mM in 10 mM PBS, pH 7.4). This protocol required approximately 2.5 hours to complete (60 reactions). Reactions were quenched using 800 µL EtOAc and vortexed for 15 seconds at max speed.

Caffeine extraction from microsomal assays. Microsomal assay samples were spun down for 3 minutes at 10,000 RPM. The top EtOAc layer (600 µL) was pipetted into an appropriately labeled GC-MS vial. The settings and parameters on the spectrometer were set following a GC-MS SIM method (Holland 2025) found in “Week 2- Caffeine Extractions from Saliva” (2024_HollandLab_Caffeine_Splitless_SIM_8min.M). The method used selected ion monitoring (SIM) at m/z 194 (Lisko et al., 2017). Converting raw abundance from GC-MS to [caffeine]. Utilizing the caffeine standard curve ($Y = 255747 \cdot X - 15451$) from “Week 3b- Graphing GC-MS Data in GraphPad Prism” (Holland 2025), raw GC-MS peak areas for all assay samples were converted to caffeine concentrations (mg/L) remaining after the time course using the equation: $[\text{caffeine}] = (\text{Peak Area} + 15451) / 255747$.

Results

Developing detection and quantification of psoralen in cilantro by GC-MS. Since microsomes originate from the collapse of the endoplasmic reticulum, they are enriched in membrane-bound enzymes, particularly cytochrome P450s such as CYP1A2, making them a robust system for probing xenobiotic metabolism (Alberts et al., 2002). Psoralen, the parent furanocoumarin from which many Apiaceae-derived analogs arise, was selected as a representative inhibitor to enable direct comparison with whole-plant extracts across this botanical family. To test whether furanocoumarins like psoralen exhibit cooperative inhibitory properties when present alongside additional phytochemicals, we treated human liver microsomes with four conditions: cilantro whole-plant extract, pure psoralen, fluvoxamine (an established CYP1A2 inhibitor), and a vehicle control. These data would ultimately contribute to a broader understanding

of diet-mediated modulation of drug metabolism. Before performing microsomal inhibition assays, we first needed to establish whether psoralen is present at detectable levels in cilantro. Prior reports identified quantifiable psoralen in celery and parsnip, two Apiaceae relatives, with targeted GC-MS in SIM mode for m/z at 187, but data for cilantro is limited (Peroutka et. al 2007). We therefore developed a GC-MS protocol to measure psoralen abundance in cilantro extract, expressed as ng/g of plant material. Using the newly developed GC-MS protocol, psoralen was successfully measured in a psoralen standard (100 µg/mL, N=1), confirming the method's suitability for detection. The compound eluted at a retention time of 6.6 min (Figure 2).

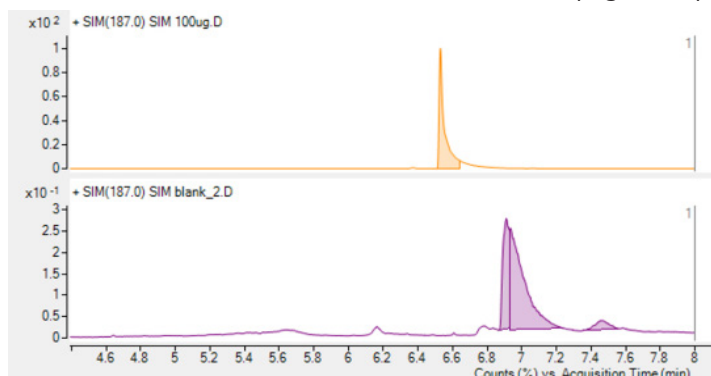


Figure 2. Detection of psoralen at 100 µg/mL on GC-MS. A psoralen standard (100 µg/mL, N=1) confirmed a new GC-MS protocol's applicability for detecting psoralen using SIM for m/z at 187. Peak was identified at a retention time of 6.6 minutes.

Quantification across concentrations 0.01, 0.1, 1, 10, and 100 µg/mL (N=1) generated a linear relationship between peak area and psoralen concentration modeled by: $\text{Peak Area} = 14361 \times [\text{psoralen}] (\mu\text{g/mL}) + 5065$ (Figure 3).

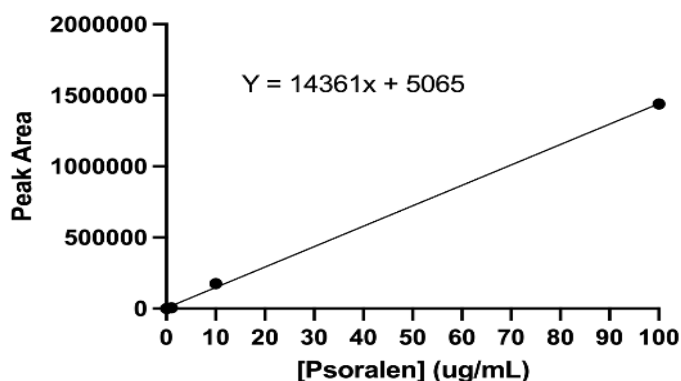


Figure 3. Standard curve relating psoralen concentration to GC-MS abundance peak area. Peak Area = 14361[psoralen] (ug/mL) + 5065 with an $R^2 = 0.999$. [Psoralen] tested were 0.01, 0.1, 1, and 100 ug/mL (N=1 for each). GC-MS analysis of cilantro whole-plant extract detected a peak with retention time 7.0 min (N=1), which, due to instrument variability, is generally accepted as sufficiently similar, corresponding to a psoralen concentration in cilantro of 716.7 ng/g (fresh weight) (Figure 4, see Appendix B for calculations).

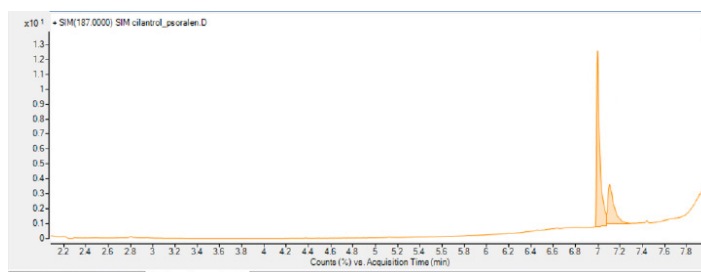


Figure 4. Detection of psoralen in cilantro whole-plant extract. A peak for m/z 187 was identified at retention time 7.0 minutes (N=1) with a peak area of 29268. This corresponded to a concentration of psoralen of 1.69 µg/mL in the sample.

Assessing CYP1A2 inhibition by cilantro extract and psoralen in human liver microsomes

We hypothesized that microsomes treated with whole-plant cilantro extract would display greater inhibition of CYP1A2 activity than those treated with pure psoralen. This expectation was based on the likelihood that cilantro contains additional furanocoumarins and other CYP1A2-active constituents capable of acting cooperatively. CYP1A2 activity was assessed by monitoring caffeine disappearance over a 50-minute time course. GC-MS analysis revealed peaks consistently at retention times near 6.5–7 minutes. Across all conditions, caffeine concentrations remained essentially unchanged (Figure 5). With overlapping flat trends for all treatments, average [caffeine] was 1.41 mg/mL. Notably, neither fluvoxamine nor the whole-plant extract treatments produced measurable inhibition relative to the vehicle.

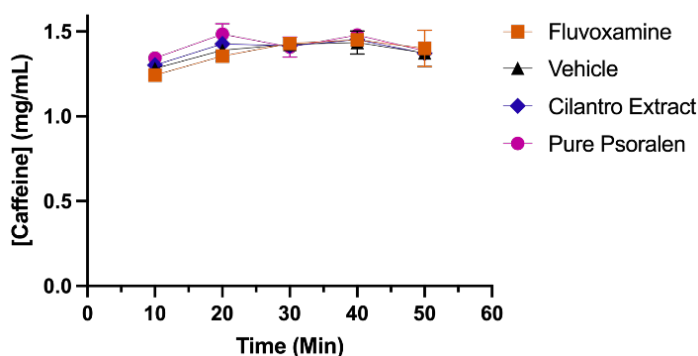


Figure 5. Metabolism of caffeine by CYP1A2. Treatments included fluvoxamine, vehicle (0.14% DMSO), a whole-plant cilantro extract, and pure psoralen. [Caffeine] (mg/mL) remained consistent throughout the time course (t=10–50 min) with an average [caffeine] of 1.41 mg/mL (range: 1.25–1.55 mg/mL). For all data points, sample sizes were N=3, with error bars representing standard deviation.

This unexpected absence of CYP1A2 inhibition contradicted our predictions and prevented

meaningful comparison of whole-plant versus pure-compound effects. As such, the data did not allow evaluation of potential synergistic interactions among furanocoumarins within cilantro extract. Instead, the uniformly flat caffeine trends suggest either insufficient inhibitory activity under the assay conditions or potential issues in assay sensitivity, enzyme activity, or experimental parameters.

Discussion

Analysis of our microsomal assay data revealed no noticeable differences in CYP1A2 activity across treatments. Notably, caffeine concentrations did not differ between our positive control (fluvoxamine) and negative control (vehicle), strongly suggesting that the assay conditions were not sufficient to generate measurable CYP1A2 activity (Figure 5). From this, the experimental outcomes are inconclusive and likely reflect technical limitations rather than true biological effects. A major limitation of our assay design was the unintended dilution of reagent concentrations. Due to last-minute adjustments to caffeine concentrations, the reaction volume was scaled to 250 μ L without proportionally scaling the concentrations of microsomes, NADPH, inhibitor treatments, or caffeine. This is reflected in our methodology. Future studies should adjust to meet the ThermoFisher protocol's call for a total reaction volume of 200 μ L. As a result, all reagents were diluted below their intended working concentrations, potentially compromising enzymatic activity and preventing measurable caffeine depletion over the assay period. NADPH concentration posed a second significant limitation, given its central role as the electron donor for CYP1A2 catalysis. The recommended concentration for CYP1A2 microsomal assays is 20 mM, yet logistical constraints required us to use NADPH at half this concentration (9.23 mM). Because NADPH is essential for cytochrome P450 catalytic activity, this reduction may have slowed reaction rates or limited the extent of caffeine metabolism. Although reducing NADPH and the total number of reactions (60 reactions rather than 72) allowed us to run more replicates, this compromise likely contributed to the low metabolic turnover observed. The preparation of the positive control also warrants consideration. While fluvoxamine is a well-established CYP1A2 inhibitor with a low IC_{50} , we used the IC_{50} concentration rather than a concentration well above it. Under ideal conditions, a positive control should produce near-complete inhibition to clearly distinguish it from the negative control; in contrast, the IC_{50} concentration is expected to only partially inhibit under optimal conditions. Although this factor alone does not explain our inconclusive

results, it should be corrected in future studies. Additionally, psoralen concentrations differed substantially between our pure psoralen treatment (0.056 nM) and our cilantro extract treatment (6.2 nM). Due to time constraints, the psoralen content in cilantro was not quantified prior to conducting the microsomal assays. Thus, we were unable to match psoralen concentrations between the purified and extract treatments. Our preliminary estimates of psoralen abundance, based on limited literature, greatly underestimated the concentration we determined via GC-MS. This further complicates the interpretation of the inhibitory conditions. Assay conditions may also have affected enzymatic activity. According to the ThermoFisher protocol, microsomal reactions should be incubated with inhibitors in a water bath at a controlled temperature, followed by incubation at 37 °C during the assay. In contrast, our reactions were carried out entirely at room temperature. Because CYP1A2 activity is temperature-dependent, conducting the reaction below physiological temperature likely slowed caffeine metabolism, contributing to the minimal turnover observed over the 50-minute time course. It is also possible that our caffeine concentration was too high for the assay window. If caffeine levels exceeded the range in which depletion could be reliably detected, the enzymes could have been saturated, thereby the reaction may have appeared static even if metabolism occurred at low rates. Optimizing substrate concentration will therefore be essential in future experiments. Finally, 50-minute microsomal assays may not have been long enough to observe a measurable decrease in caffeine concentration. It should be noted that previous studies demonstrated CYP1A2-mediated caffeine metabolism and inhibition in as little as 10 minutes (Alehaideb et al., 2021). In future studies, an initial single-time-point assay should be performed to verify CYP1A2 inhibition before committing to a larger time course experiment. Despite these limitations, our study successfully established a robust GC-MS method for quantifying psoralen in cilantro. Published measurements of psoralen content in cilantro are limited, and our approach enabled rapid determination of psoralen abundance, providing a foundation for future work examining the contributions of individual furanocoumarins to CYP1A2 inhibition. Our measured concentration of 716.7 ng/g was nearly three times higher than the reported value of 258.2 ng/g (Melough et al., 2017). That study employed an Agilent QuEChERS kit, originally developed for pesticide residue analysis,

which incorporates multiple wash steps that may reduce psoralen recovery. In contrast, our more simplified extraction method involved blending cilantro with water (without wash steps), followed by a single EtOAc extraction, which may have resulted in higher apparent yields. They also used an internal standard to calibrate their measurements, and differences in calibration strategy may further account for the difference between our measured concentrations. Moving forward, several improvements should be implemented. First, optimizing the psoralen extraction protocol is a priority to improve recovery and enable more direct comparison with published values. Our initial extraction (200 μ L extract: 800 μ L EtOAc) produced an inconclusive signal on the GC-MS, prompting us to shift to a 45 mL extract: 5 mL EtOAc ratio, which yielded reliable GC-MS detection. While this allowed for quantification, alternative extract-to-solvent ratios should be explored to maximize recovery and reproducibility. Second, isolated testing of additional furanocoumarins and flavonoids is warranted, because although psoralen is the parent molecule of many Apiaceae furanocoumarins, it may not be the primary contributor to CYP1A2 inhibition. Other constituents of cilantro may exhibit stronger inhibitory activity, and comparing their effects to whole-plant extract will help clarify potential cooperative interactions. These improvements, in parallel with greater replicates of the conditions outlined in this study, should yield clarifying results on the nature of psoralen inhibition of CYP1A2. Lastly, once data is acquired, statistical analysis should be performed. Using GraphPad Prism version 10.4.1, the extra-sum-of-squares between treatments should be calculated and analyzed for changes in the half-life of caffeine metabolism. This significance (or lack thereof) would be interpreted in the context of biological processes and physiological conditions. In summary, although our microsomal assays yielded inconclusive results, the study highlights several methodological refinements necessary for accurately assessing CYP1A2 inhibition by natural products. Our psoralen quantification method provides an analytically simple yet effective tool for future work, and continued optimization of assay conditions and extraction procedures will help deepen our understanding of food-drug interactions involving furanocoumarins.

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Appendix B. Sample Calculation of [psoralen] in cilantro.

Cilantro concentration in extraction sample

Initial extraction: 30 g cilantro / 150 mL H₂O = 0.20 g/mL (cilantro/H₂O)

For 45 mL of extract: 45 mL * 0.20 g/mL = 9 g cilantro

Psoralen concentration in EtOAc layer

Standard curve equation: $Y = 14361 * X + 5065$

Peak area from extraction: 29267.6

Calculated [psoralen]: $(29267.6 - 5065) / 14361$
= 1.69 µg / mL (Psoralen/EtOAc)

Total psoralen in EtOAc layer

5 mL EtOAc * 1.69 µg/mL = 6.45 µg psoralen

Normalized psoralen content per cilantro mass

6.45 µg psoralen / 9 g cilantro = 0.72 µg/g cilantro * 1000 ng/ug

Equivalent to **716.7 ng/g** (Psoralen/cilantro)

Appendix A. Formation of human liver microsomes.

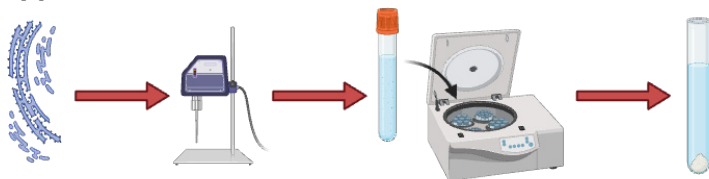


Diagram showing the formation of microsomes and centrifugation (created on biorender).